Bridging Learning Outcomes and Module Specifications: GenAI Framework for Curriculum Development in CS Education

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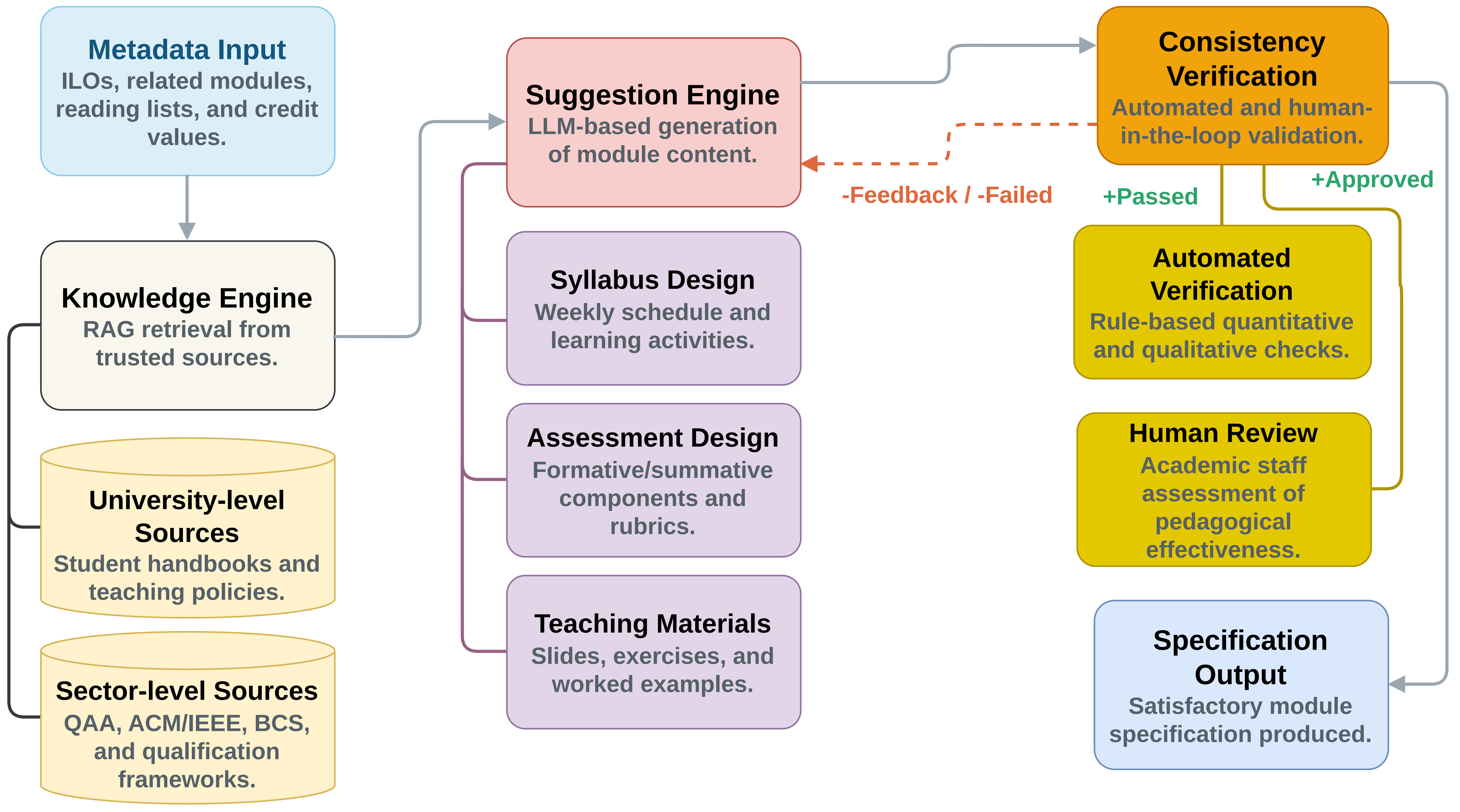
Background

Context. Recent developments in GenAI have enabled the production of high-quality content across education and computing sectors. This opens up opportunities to modernise the curriculum development and learning experience among tutors and students.

Idea. This work proposes a four-stage framework for curriculum development of computer science modules: (1) Metadata Input, (2) Knowledge Engine, (3) Suggestion Engine, and (4) Consistency Verification.

Proposed Framework



Conclusions and Future Work

Summary. An end-to-end GenAI-driven pipeline for automated suggestion of computing module specifications, integrating trusted academic and accreditation sources with a hybrid verification mechanism to ensure regulatory compliance, and pedagogical validity. **Next Steps.** (i) Incorporating international institutional constraints. (ii) Extending support to joint-institutional curriculum development.

Thanks

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